

Software Requirements Specification (SRS)
Project: Smart Irrigation Management System

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1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the Smart Irrigation Management System. The system will enable farmers and farm administrators to remotely monitor farms, automate irrigation, manage crops, and generate reports using IoT sensor data.

1.2 Project Scope

The Smart Irrigation Management System aims to:

- Monitor live sensor readings.
- Automate irrigation based on soil moisture.
- Reduce water consumption.
- Improve crop yield.
- Provide reports and analytics.
- Enable remote farm management.

1.3 Intended Users

- Farm Owners
- Supervisors
- Caretakers
- System Administrators

2. Business Objective

The client currently operates irrigation manually, leading to:

- Excessive water usage
- Delayed irrigation
- Crop damage
- High operational cost

The proposed solution should:

- Automate irrigation.
- Reduce manual intervention.
- Provide real-time farm monitoring.
- Improve productivity.
- Reduce water wastage.

3. Functional Requirements

Module 1 - Authentication

FR-1

The system shall allow users to log in using their email address and password.

FR-2

The system shall support password reset.

FR-3

The system shall support secure JWT authentication.

Module 2 - User Management

FR-4

The administrator shall be able to:

Create users

Edit users

Delete users

Assign roles

FR-5

Supported roles:

Admin

Owner

Supervisor

Caretaker

Module 3 - Farm Management

FR-6

Users shall be able to:

Add farms

Edit farms

Delete farms

View farm details

Each farm should contain:

Farm Name

Address

GPS Coordinates

Area

Crop Details

Module 4 - Sensor Management

The system shall support:

Soil Moisture Sensors
Temperature Sensors
Humidity Sensors
pH Sensors
EC Sensors
Dashboard shall display:
Current Moisture
Temperature
Humidity
Battery Level
Last Updated Time
Sensor Status
Module 5 - Irrigation Automation
Automatic Irrigation

The system shall automatically start irrigation when:

Current Moisture less than Lower Threshold

The system shall automatically stop irrigation when:

Current Moisture greater or equal to Upper Threshold

Users shall also be able to:

Start irrigation manually
Stop irrigation manually
Module 6 - Dashboard

The Owner Dashboard shall display:

Total Farms
Total Sensors
Active Sensors
Offline Sensors
Total Plants
Total Valves
Total Harvest in Kg
Total Harvest in Pieces
Pending Tasks
Delayed Tasks
Observation Count
Water Consumption
Irrigation Status
Module 7 - Crop Management

Users shall be able to:

- Add Crop
- Edit Crop
- Delete Crop
- View Crop History

Each crop contains:

- Crop Name
- Variety
- Planting Date
- Harvest Date
- Expected Yield
- Module 8 - Yield Management

The system shall:

- Calculate expected yield.
- Calculate actual yield.
- Show variance.
- Display monthly yield trends.

Business Rule:

- Negative variance should display as 0.
- Module 9 - Harvest Management

Users shall record:

- Harvest Date
- Quantity
- Unit

Supported Units:

- Kilograms
- Pieces

Dashboard shall display totals separately.

Module 10 - Task Management

Users shall:

- Create Tasks
- Update Tasks
- Assign Tasks

Complete Tasks

Tasks shall be sorted by:

Due Date instead of Created Date

Module 11 - Notifications

The system shall notify users when:

Sensor goes offline
Battery becomes low
Irrigation starts
Irrigation stops
Task is overdue

Notifications delivered via Mobile and Email.

Module 12 - Reports

The system shall generate:

Daily Report containing Irrigation Summary, Water Usage, Sensor Status

Weekly Report containing Crop Health, Water Consumption, Harvest Summary

Monthly Report containing Farm Performance, Yield Analysis, Sensor Uptime, Task Completion

Reports shall be exportable as PDF and Excel.

4. Non-Functional Requirements

Performance

Dashboard shall load within 3 seconds.
APIs shall respond within 2 seconds under normal load.

Availability

System uptime shall be 99.9 percent.

Security

JWT Authentication, HTTPS, Encrypted passwords, Role-based access control, Audit logging.

Scalability

The system shall support 10,000 farms, 100,000 sensors, 50,000 concurrent users.

Reliability

Daily automatic backups. Recovery Time Objective less than 2 hours.

Browser Support

Chrome, Edge, Firefox, Safari.

5. User Roles

Admin has full system access.

Owner manages farms, users, reports.

Supervisor assigns tasks, monitors farms.

Caretaker views assigned farms, completes tasks.

6. Reports

Dashboard Report displays Active Farms, Total Sensors, Water Usage, Harvest, Tasks, Observation Count.

Yield Report displays Expected Yield, Actual Yield, Variance, Monthly Trends.

Irrigation Report displays Irrigation Duration, Water Used, Pump Status, Valve Activity.

Sensor Health Report displays Online Sensors, Offline Sensors, Battery Status, Communication Errors.

Task Report displays Pending Tasks, Completed Tasks, Delayed Tasks, Assigned Users.

7. Acceptance Criteria

The project shall be considered complete when users can securely log in, farms can be created and managed, sensors display real-time data, automatic irrigation works based on configured thresholds, manual irrigation controls function correctly, dashboard displays all required metrics, observation count is shown on the Owner Dashboard, yield variance never displays a negative value, tasks are sorted by Due Date, reports can be exported as PDF and Excel, notifications are sent successfully, APIs meet the defined performance requirements, and the system passes User Acceptance Testing with no critical defects.

8. Assumptions

Sensors transmit valid data at configured intervals. Internet connectivity is available at farm locations. Users have appropriate permissions assigned. Third-party services such as email and SMS are operational.

9. Future Enhancements

AI-based irrigation recommendations. Weather forecast integration. Pest and disease prediction.

Mobile application for Android and iOS. Voice assistant support. Multi-language interface.
Predictive yield analytics using machine learning.